

EXPONENTIAL OF DELTA EXPONENT ENERGY APPLICATION

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ABSTRACT

This paper presents the Exponential of Delta Exponent Energy Application system based on Rumus Perpangkatan Universal 4.0. The framework incorporates multi-disciplinary IoT telemetry, business financial ROI analytics, FinTech QRIS micro-metering payment gateways, and embedded CAD circuit schematic blueprints for smart micro-grid energy converters.

1. MATHEMATICAL FORMULATION & ENERGY DERIVATION

$E_{\text{delta}} = \text{Integral}_0^t [(x-y)^n * e^{-\alpha * \tau} + \text{Integral}_0^1 x^x dx] d(\tau)$
Newton Binomial Expansion: $(x-y)^n = \sum_{k=0}^n C(n, k) x^{n-k} (-1)^k y^k$
Sophomore's Dream Constant: $\text{Integral}_0^1 x^x dx = 0.783430510712134$

2. MULTI-DISCIPLINE EMBEDDED CIRCUIT & IOT SYSTEM

- Microcontroller MCU: STM32F4 / ESP32-S3 Dual-Core 240MHz Engine
- Sensor Transducers: ACS712-30A Hall Current Sensor & B25 Voltage Transducer
- Power Stage: High-Frequency Dynamic MOSFET Switching Array
- FinTech Gateway: Dynamic QRIS Micro-Payment Settlement & Telemetry Stream

3. SCOPUS Q1 BIBTEX CITATION & IEEE COMPLIANCE

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